A Short History of the Printing Press

From woodblocks to digital printing

Before the Printing Press

For most of history, books were copied by hand. In medieval Europe the work was done mostly by monks in monasteries, who spent months copying a single volume with quill and ink. A book could take a year to finish, so each one was extremely expensive, and only churches, universities and rich families could afford them.

Printing itself was invented much earlier in East Asia. In China, craftsmen carved whole pages of text into wooden blocks, covered them with ink, and pressed paper onto them. This method, called woodblock printing, was already in use more than a thousand years ago. It worked well for pages that would be printed many times, but a new block had to be carved for every page.

Around the year 1040, during the Song dynasty, a Chinese craftsman named Bi Sheng invented movable type. He made individual characters out of baked clay that could be arranged into a page and then reused for a different text. The idea was clever, but the Chinese writing system uses thousands of different characters, which made the method awkward to use.

In Korea, printers took the idea further by casting movable type from metal. The Jikji, a Buddhist text printed in 1377, is the oldest surviving book printed with movable metal type. It was made about seventy years before the famous work of Johannes Gutenberg in Europe, though there is no evidence that the two traditions were connected.

By the middle of the fifteenth century Europe was ready for a change. Paper had become cheaper, literacy was growing in the towns, and merchants, students and priests all wanted more books than the scribes could provide.

Johannes Gutenberg and the Mainz Workshop

Johannes Gutenberg was born in the German city of Mainz around the year 1400. He trained as a goldsmith, and that skill proved crucial. Around 1450 he began to develop a complete system for printing with movable metal type, and he set up a workshop in Mainz.

Gutenberg's genius was to combine several technologies into one practical process. He designed a hand mould that could cast hundreds of identical metal letters quickly, using an alloy of lead, tin and antimony that melted easily but hardened into durable type. He created an oil-based ink that stuck to metal, unlike the water-based inks used for wooden blocks. And he adapted a screw press, similar to those used for making wine and olive oil, to press the paper evenly against the type.

The work needed money, and Gutenberg borrowed heavily from a wealthy lender named Johann Fust. His most ambitious project was the Bible in Latin, printed in about 1455. The Gutenberg Bible has 42 lines on each page, and it was printed in two volumes. It is commonly estimated that around 180 copies were made, some on paper and some on vellum. Fewer than fifty copies survive today, and they are among the most valuable printed books in the world.

Gutenberg himself did not become rich. Around 1455 Fust took him to court over the unpaid loans, and the judges ruled in Fust's favour. Fust gained control of much of the equipment and went on to run a successful printing business. Gutenberg died in Mainz in 1468, and his invention was already spreading beyond the city.

The significance of his work was not that it was the first printing, but that it made printing fast, repeatable and cheap enough to change the economics of the book.

Printing Spreads Across Europe

After the Mainz workshop was disturbed by a conflict in 1462, trained printers left the city and carried the craft with them. Presses soon appeared in Italy, France, the Netherlands, Spain and England. Within a few decades there were printing shops in hundreds of European towns.

Venice became one of the great centres of the new industry. The printer Aldus Manutius worked there from the 1490s. He published fine editions of classical Greek and Latin authors, introduced a slanted style of type that we now call italic, and printed small portable books that could fit in a pocket. These ideas made reading more mobile and personal.

William Caxton, an English merchant, learned printing during his time in Bruges and Cologne. In 1476 he set up the first printing press in England at Westminster, near London. He printed more than one hundred titles, including The Canterbury Tales by Geoffrey Chaucer.

Books printed before the year 1501 are called incunabula, from a Latin word meaning the cradle, because they belong to the infancy of printing. Historians have recorded tens of thousands of different editions from this early period, and many survive in libraries today.

Printing changed the world of ideas. When Martin Luther published his Ninety-Five Theses in 1517, printers copied and spread them across Germany within weeks. The Reformation, the Scientific Revolution and the growth of public education all depended on the ability to share the same text with thousands of readers.

Printing in the Romanian Lands

The printing press reached the Romanian lands early in the sixteenth century. The first book printed on the territory of present-day Romania was a liturgical book, the Liturghier, printed in 1508 in Targoviste, then the capital of Wallachia. It was produced by a monk named Macarie, who had learned his craft abroad, and it was written in Church Slavonic, the language used in church services at the time.

Macarie later printed two more books in Targoviste, an Octoechos in 1510 and a Gospel book in 1512. Together these works helped establish the Wallachian court as a centre of learning and showed that printing was no longer limited to Western Europe.

Printing in Romanian itself came a little later. In the second half of the sixteenth century, a deacon named Coresi ran a printing workshop in Brasov, in Transylvania. In the 1560s he printed religious books in Romanian, including some of the earliest surviving texts in the language. His work encouraged the use of Romanian in church and in writing.

In the following centuries, printing houses were opened in monasteries and cities across Wallachia, Moldavia and Transylvania. Many of the books they produced were religious texts, but over time they also included laws, school books and chronicles.

The spread of printing in these lands was an important step toward a shared written culture, and it supported later movements for education and national awareness.

From Steam Presses to Digital Printing

For almost four hundred years the basic design of the printing press changed very little. Printers still set each letter by hand and pulled the press lever by hand. The first major leap came in the early nineteenth century, when the German inventor Friedrich Koenig built a steam-powered press. In 1814 the London newspaper The Times began to be printed on Koenig's machine, which could produce far more sheets per hour than a hand press.

Setting type was still slow until the Linotype machine appeared in 1886. Invented by Ottmar Mergenthaler, it had a keyboard, and the operator typed a line of text that the machine cast as a single metal slug. Linotype machines made it possible to produce daily newspapers with large numbers of pages.

In the twentieth century offset printing became the main commercial method. In this process the image is transferred from a plate to a rubber blanket and then onto the paper, giving sharp results at high speed. Photographic typesetting, and later computers, replaced metal type almost completely by the end of the century.

Today digital printing allows a document to go straight from a computer file to paper, with no plate at all. This makes it economical to print a single copy or a small run, and it has helped create new fields such as print on demand and desktop publishing.

More than five hundred years after Gutenberg, the central idea remains the same: produce text once, then copy it quickly and accurately for many readers. Even e-books and websites continue the tradition of making information widely available.